

Sustainability aware asset management

Lecture 3: Black-Litterman Approach and Risk Budgeting

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Objectives of the Lecture

Previous lecture:

- The plug-in optimizer reacts strongly to small changes in $\hat{\mu}$.
- Empirically: turnover, concentration, and unstable exposures can dominate performance.

We can rely on **two complementary solution families**:

- **Fix $\hat{\mu}$** : discipline expected returns via structure/prior (Black-Litterman).
- **Avoid $\hat{\mu}$** : build portfolios from risk only (Risk Budgeting).

In both cases, you still need a stable risk model $\hat{\Sigma}$ and implementable constraints

- **Black and Litterman** (1992), *Global Portfolio Optimization*, Financial Analysts Journal 48, 28–43.

⇒ Seminal description of the Black-Litterman

- **Idzorek** (2004), *A step-by-step guide to the Black-Litterman model*, working paper.

⇒ Guide to Black-Litterman approach

- **Maillard, Roncalli, and Teiletche** (2010), *The Properties of Equally Weighted Risk Contribution Portfolios*, Journal of Portfolio Management, 36(4), 60–70.

⇒ Presentation of the equally-weighted risk contribution portfolio.

Objectives of the lecture

- **Black-Litterman Approach**
- Risk Budgeting

Black-Litterman Approach

Managers may have a passive benchmark (say, the market) but use their own views for designing an active strategy (tactical asset allocation).

Managers may be reluctant to use historical average returns for their neutral view because

- sample means are too noisy
- sample means are not neutral, as they favor past winners.

Instead, priors (“neutral” views) about risk premia are derived from market equilibrium.

The **Black-Litterman approach** combines:

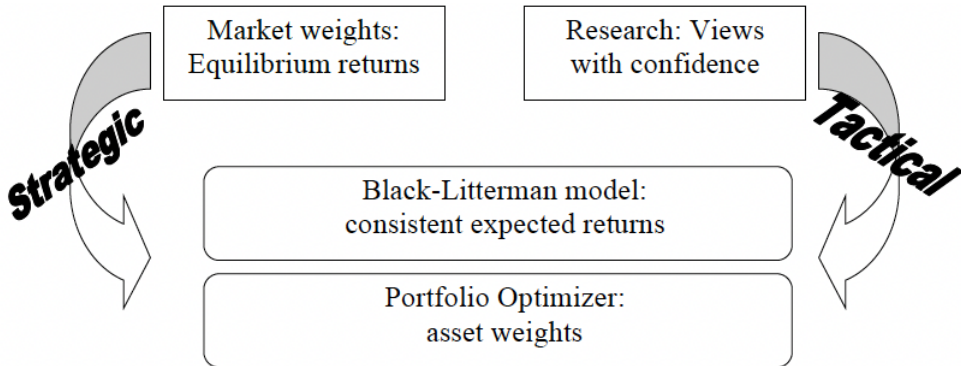
- the managers’ subjective views regarding the expected returns
- with the market equilibrium expected returns (prior distribution)
- to form a new estimate of expected returns (posterior distribution)

This approach is an application of the mixed estimation approach: it attempts to improve estimates of mean returns by combining information from data with the investor’s priors.

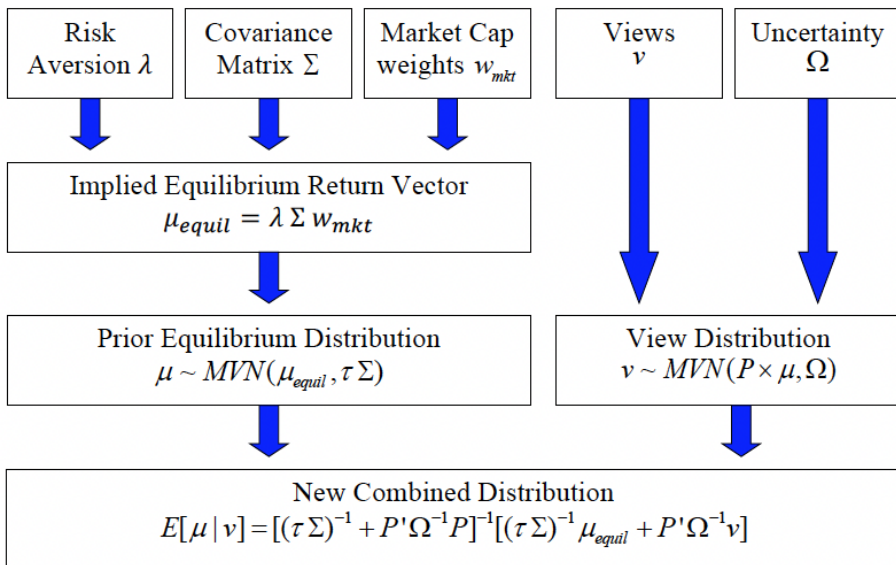
The Logic

Intuition:

- if you do not have views, you should hold the market portfolio
- your views will tilt the final weights away from the market portfolio, the degree to which depending on how confident you are about your views.



The Logic



Step 1: CAPM Equilibrium Risk Premia

Excess return distribution: Excess returns are drawn from an i.i.d. normal distribution:

$$\mathbf{R}_{t+1} - R_{f,t} \sim \mathcal{N}(\boldsymbol{\mu}^e, \boldsymbol{\Sigma}),$$

with $\boldsymbol{\mu}^e$: risk premium (or expected excess return).

In equilibrium, all investors hold the market portfolio.

Equilibrium risk premia: Premia that induce a risk-averse investor to hold all available assets in proportion to their current market capitalizations.

With mean-variance utility (Tobin's approach): vector of market cap weights $\boldsymbol{\alpha}_{mkt}$,

$$\boldsymbol{\alpha}_{mkt} = \frac{1}{\lambda} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_{equil}^e \quad \Rightarrow \quad \boldsymbol{\mu}_{equil}^e = \lambda \boldsymbol{\Sigma} \boldsymbol{\alpha}_{mkt}.$$

Step 1: CAPM Equilibrium Risk Premia – What value for λ ?

What should be the risk aversion parameter to use at the equilibrium?

Suggestion: The average market risk version

$$\alpha'_{mkt} \mu^e_{equil} = \lambda \alpha'_{mkt} \Sigma \alpha_{mkt}$$

so that

$$\lambda = \frac{\alpha'_{mkt} \mu^e_{equil}}{\alpha'_{mkt} \Sigma \alpha_{mkt}} = \frac{\mathbb{E}[R_m] - R_f}{\text{Var}(R_m)} = \frac{SR_m}{\sqrt{\text{Var}(R_m)}}$$

λ is often calibrated to the historical Sharpe ratio (e.g., $\lambda = 2.5$ or 5).

With this λ , equilibrium risk premia become:

$$\mu^e_{equil} = SR_m \frac{\Sigma \alpha_{mkt}}{\sqrt{\alpha'_{mkt} \Sigma \alpha_{mkt}}}$$

Step 1: CAPM Equilibrium Risk Premia – Illustration

Monthly returns for 6 portfolios of all AMEX, NASDAQ, and NYSE stocks sorted by their market capitalization and book-to-market ratio from 1983 to 2003. (Source: Brandt, 2010)

Descriptive statistics on portfolio returns

Size	Book to market	Sample risk premia (%)	Volatility (%)	Correlations				
Small	Low	5.61	24.56	1				
Small	Medium	12.75	17.01	0.926	1			
Small	High	14.36	16.46	0.859	0.966	1		
Big	Low	9.72	17.07	0.784	0.763	0.711	1	
Big	Medium	10.59	15.05	0.643	0.768	0.763	0.847	1
Big	High	10.44	13.89	0.555	0.698	0.735	0.753	0.913

Remark: **Sample risk premia** are based on the sample means of portfolio returns, so there are extremely noisy. The Small-Low portfolio has the lowest risk premium while it has the highest volatility. **Equilibrium risk premia** are biased estimates of the risk premia (because they are based on a model), but they are more precisely estimated.

Step 1: CAPM Equilibrium Risk Premia – Illustration

Equilibrium risk premia implied by market capitalization weights

Size	Book to market	Market weight (%)	Equilibrium risk premia (%)				Historical risk premia (%)
			$\lambda = 1$	$\lambda = 2.5$	$\lambda = 5$	$\lambda = 7.5$	
Small	Low	2.89	3.07	7.69	15.37	23.06	5.61
Small	Medium	3.89	2.21	5.52	11.03	16.55	12.75
Small	High	2.21	2.04	5.11	10.22	15.33	14.36
Big	Low	59.07	2.62	6.55	13.10	19.64	9.72
Big	Medium	23.26	2.18	5.44	10.88	16.32	10.59
Big	High	8.60	1.97	4.91	9.83	14.74	10.44

Comments:

- The levels of the risk premia depend on the level of risk aversion, which therefore needs to be calibrated before using the results in the mixed estimator.
- The implied equilibrium risk premia are quite different from the empirical risk premia. The two sets of risk premia are negatively correlated in the cross-section (a correlation coefficient of -0.83).

Step 2: Priors and Subjective Views

Prior distribution

Benchmark beliefs about μ^e are centered about the equilibrium risk premia:

$$\mu^e \sim \mathcal{N}(\mu_{equil}^e, \tau \Sigma),$$

where τ is scaling the return covariance matrix (e.g., $\tau = 1/T$).

Subjective Views

Suppose now that the investor also has K subjective views:

$$\mathbf{P}\mu^e = \mathbf{q} + \boldsymbol{\eta} \quad \text{with} \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$$

where

- $\mathbf{P}$ is a $K \times N$ matrix combining returns into portfolios,
- $\mathbf{q}$ is a $K \times 1$ vector of view returns,
- $\boldsymbol{\Omega}$ is a $K \times K$ forecast error covariance matrix (often diagonal, i.e. no cross-information on views)

Step 3: Combining Equilibrium and Views

We have

$$\boldsymbol{\mu}^e \sim \mathcal{N}(\boldsymbol{\mu}_{equil}^e, \tau \boldsymbol{\Sigma}) \quad (\text{equilibrium premia})$$

$$\mathbf{P}\boldsymbol{\mu}^e = \mathbf{q} + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}) \quad (\text{views}).$$

Define the posterior (mixed) estimate:

$$\boldsymbol{\mu}_{BL}^e = \left[(\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}' \boldsymbol{\Omega}^{-1} \mathbf{P} \right]^{-1} \left[(\tau \boldsymbol{\Sigma})^{-1} \boldsymbol{\mu}_{equil}^e + \mathbf{P}' \boldsymbol{\Omega}^{-1} \mathbf{q} \right].$$

Distribution of the mixed risk premia:

$$\boldsymbol{\mu}^e \mid \text{views} \sim \mathcal{N}\left(\boldsymbol{\mu}_{BL}^e, \left[(\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}' \boldsymbol{\Omega}^{-1} \mathbf{P} \right]^{-1}\right).$$

Limit cases:

- no view: $\mathbf{P}$ empty $\Rightarrow \boldsymbol{\mu}_{BL}^e = \boldsymbol{\mu}_{equil}^e$
- no uncertainty on views: $\boldsymbol{\Omega} \rightarrow \mathbf{0} \Rightarrow \mathbf{P}\boldsymbol{\mu}_{BL}^e = \mathbf{q}$
- no confidence: $\boldsymbol{\Omega} \rightarrow \infty \Rightarrow \boldsymbol{\mu}_{BL}^e = \boldsymbol{\mu}_{equil}^e$

Step 3: Combining Equilibrium and Views

As expected excess returns μ^e are random variables in the Black-Litterman model, the predictive distribution of the excess returns is not simply $\mathcal{N}(\mu_{BL}, \Sigma)$.

Predictive return distribution with known Σ :

$$\mathbf{R}_{t+1}^e \mid \text{views} \sim \mathcal{N}(\mathbb{E}[\mathbf{R}_{t+1}^e \mid \text{views}], \text{Var}[\mathbf{R}_{t+1}^e \mid \text{views}])$$

$$\sim \mathcal{N}(\mathbb{E}[\mu^e \mid \text{views}], \Sigma + \text{Var}[\mu^e \mid \text{views}])$$

$$\sim \mathcal{N}(\mu_{BL}, \Sigma_{BL})$$

where

$$\Sigma_{BL} = \Sigma + \left[(\tau \Sigma)^{-1} + \mathbf{P}' \Omega^{-1} \mathbf{P} \right]^{-1}$$

Step 4: Weight Optimization

Given the predictive mean μ_{BL} and covariance matrix Σ_{BL} , we can now construct the optimal portfolio using a standard mean-variance optimization method.

Assume the maximization problem

$$\max_{\{\alpha\}} \alpha' \mu_{BL} - \frac{\lambda}{2} \alpha' \Sigma_{BL} \alpha.$$

We obtain the optimal weights:

$$\alpha^* = \frac{1}{\lambda} \Sigma_{BL}^{-1} \mu_{BL}.$$

Remark: When $\mu_{BL} = \pi = \lambda \Sigma_{BL} \alpha_{mkt}$ (when no views), we find that: $\alpha^* = \alpha_{mkt}$.

Illustration

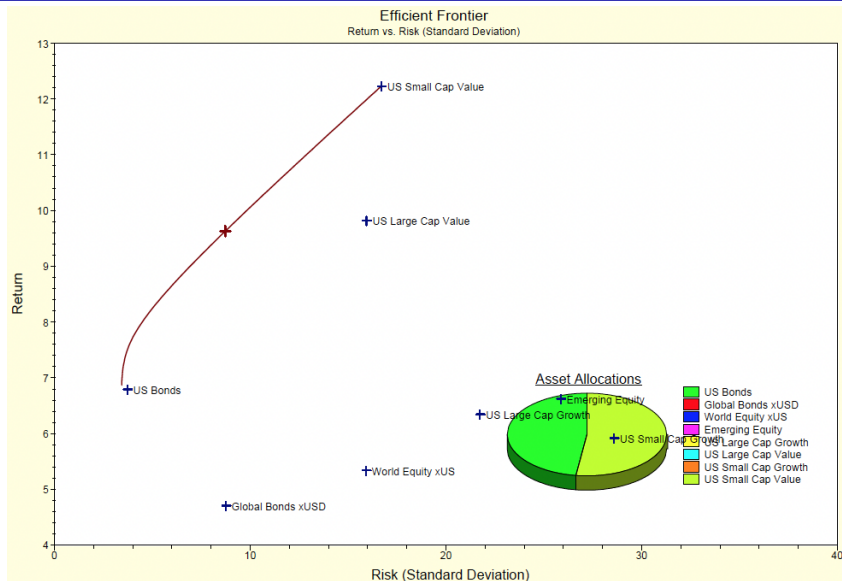
Idzorek (2004), A step-by-step guide to the Black-Litterman Model.

International markets.

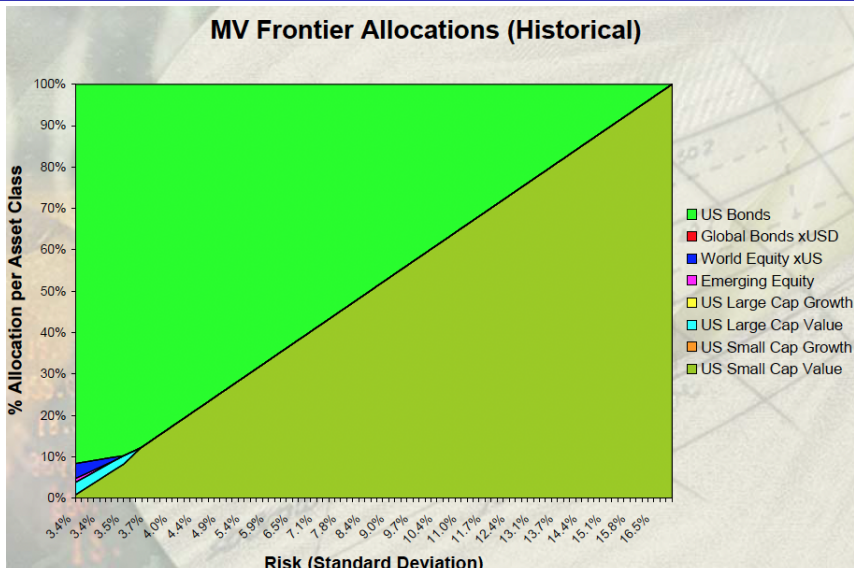
Descriptive statistics on asset classes returns

Asset classes	Historical return (%)	Market cap (%)	Risk premium (%)	Risk-free rate (%)	Total implied return (%)
US bonds	6.8	20.2	0.1	4.0	4.1
Global bonds xUSD	4.7	27.9	1.0	4.0	5.0
World Equity xUSD	5.4	22.2	4.0	4.0	8.0
Emerging Equity	6.5	2.33	5.4	4.0	9.4
US Large cap Growth	6.3	12.6	5.1	4.0	9.1
US Large cap Value	9.8	12.6	3.7	4.0	7.7
US Small cap Growth	6.0	1.1	6.1	4.0	10.1
US Small cap Value	12.2	1.1	3.5	4.0	7.5

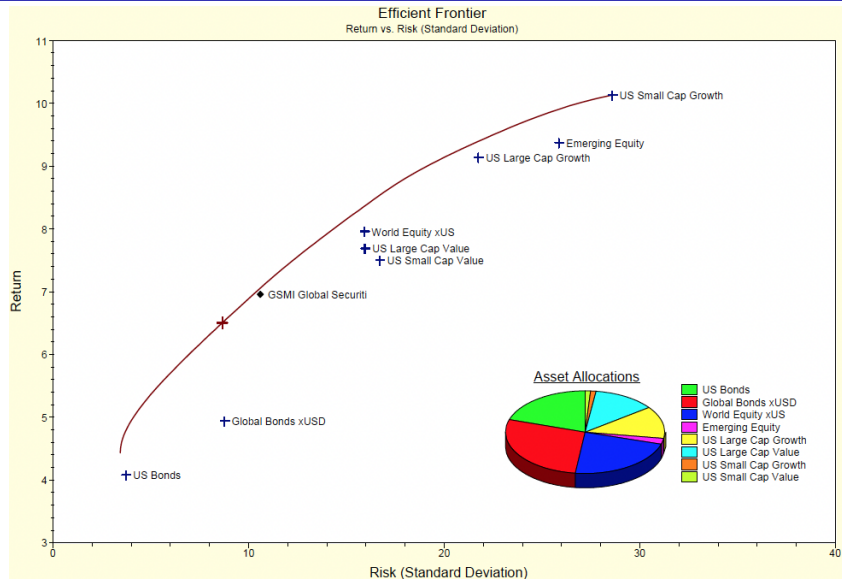
Efficient frontier based on Historical returns and covariance matrix



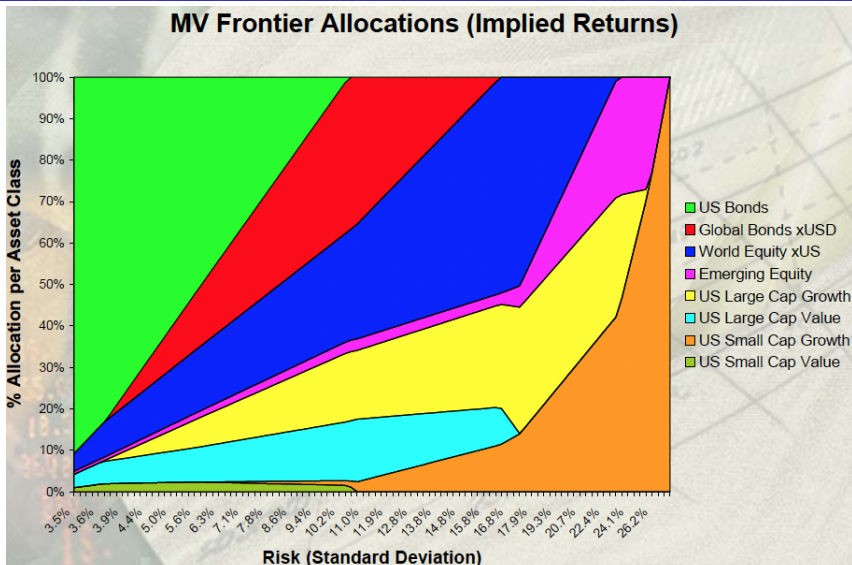
Optimal weights based on Historical returns and covariance matrix



Efficient frontier based on implied returns and historical covariance matrix



Optimal weights based on implied returns and historical covariance matrix



Illustration

Incorporating a view: The US equity will outperform World equity xUS by 150 bp (75% confidence)

Ex ante situation:

- Implied return of World equity xUS = 8%
- Implied return of US equity:

$$\frac{(9.1\% \times 12.6\%) + (7.7\% \times 12.6\%) + (10.1\% \times 1.1\%) + (10.1\% \times 1.1\%)}{7.5\% + 12.6\% + 1.1\% + 1.1\%}$$

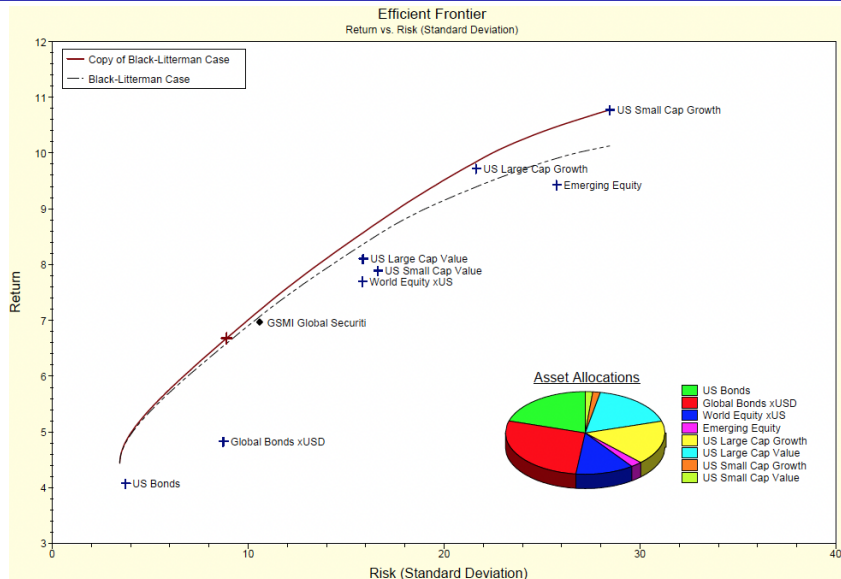
The implied return of World equity is 40 bp below the implied return of US equity

Define the view as (illustrative form):

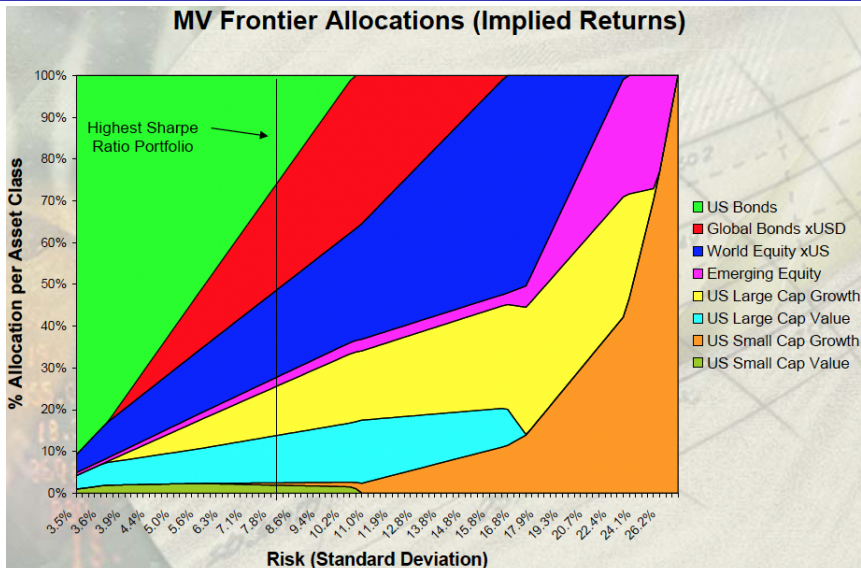
$$\underbrace{\begin{bmatrix} 0 & 0 & -1 & 0 & 1/4 & 1/4 & 1/4 & 1/4 \end{bmatrix}}_{\mathbf{P}} \boldsymbol{\mu} = \underbrace{0.015}_{q},$$

with confidence matrix will be $\boldsymbol{\Omega} = \tau \boldsymbol{\Sigma}$ and confidence level $\tau = 0.75$ (Idzorek's calibration).

Efficient frontier based on BL with views



Optimal weights based on implied returns with views



Objectives of the Lecture

- Black-Litterman Approach
- **Risk Budgeting**

Risk Budgeting

Risk Budgeting has been proposed as an alternative to the usual portfolio optimization. Its objective is to control asset risk contributions to total risk.

In a standard stock-bond (40%-60%) portfolio, stocks are performance assets, while bonds are usually considered as a diversifying, risk-reducing asset (deleveraging the equity portfolio). The weight budget (40%-60%) does not reflect the risk budget.

Risk Budgeting is the process of:

- decomposing the aggregate risk of a portfolio (or even of the entire investment process) into its constituents (risk factors or drivers) on a quantitative basis
- setting risk limits (risk budgets) to each asset class (or even single asset), factor and/or investment manager (ex ante)
- allocating assets in compliance with risk budgets

Risk Budgeting does not rely on expected returns

Risk Contribution

We define the risk budget using the volatility: $\sigma_p(\alpha) = \sqrt{\alpha' \Sigma \alpha}$

How would the portfolio volatility $\sigma_p(\alpha)$ change if we increase weight of a particular asset?
This is the **marginal contribution to risk (MCR)**:

$$\text{MCR} = \frac{\partial \sigma_p}{\partial \alpha} = \frac{\partial \sqrt{\alpha' \Sigma \alpha}}{\partial \alpha} = \frac{\Sigma \alpha}{\sqrt{\alpha' \Sigma \alpha}}$$

or, for asset i :

$$\text{MCR}_i = \frac{\partial \sigma_p}{\partial \alpha_i} = \frac{\alpha_i \sigma_{ii} + \sum_{j \neq i}^N \alpha_j \sigma_{ij}}{\sigma_p} = \sum_{j=1}^N \frac{\alpha_j \sigma_{ij}}{\sigma_p} = \frac{\sigma_{ip}}{\sigma_p} = \frac{\sigma_{ip}}{\sigma_p^2} \sigma_p = \beta_{ip} \sigma_p$$

where $\sigma_{ip} = \text{Cov}[R_i, R_p] = \alpha_i \sigma_{ii} + \sum_{j \neq i}^N \alpha_j \sigma_{ij}$ is the covariance of asset i with the portfolio.

Risk Contribution

Then, the portfolio volatility can be rewritten as the weighted average of the MCR:

$$\sigma_p = \sum_{i=1}^N \frac{\alpha_i \sigma_{ip}}{\sigma_p} = \sum_{i=1}^N \alpha_i \text{MCR}_i.$$

We also define the **risk contribution**:

$$\text{RC} = \alpha \frac{\partial \sigma_p}{\partial \alpha}$$

or, for asset i :

$$\text{RC}_i = \sigma_i(\alpha) = \alpha_i \frac{\partial \sigma_p}{\partial \alpha_i} = \alpha_i \beta_i \sigma_p \quad \text{and} \quad \text{RC}_i = \alpha_i \frac{\partial \sigma_p}{\partial \alpha_i} = \alpha_i \frac{(\mathbf{\Sigma} \alpha)_i}{\sigma_p}$$

so that

$$\sum_{i=1}^N \text{RC}_i = \sigma_p$$

The sum of individual risk contributions gives the portfolio volatility.

Illustration

We consider three assets, with expected returns equal to zero and volatilities are equal to $\sigma = (0.3, 0.2, 0.15)'$. The correlation matrix of asset returns is:

$$\rho = \begin{bmatrix} 1 & 0.8 & 0.5 \\ 0.8 & 1 & 0.3 \\ 0.5 & 0.3 & 1 \end{bmatrix}$$

We consider the portfolio weights $\alpha = (0.5, 0.2, 0.3)'$

The covariance matrix is:

$$\Sigma = \text{diag}(\sigma) \rho \text{diag}(\sigma) = \begin{bmatrix} 9.00 & 4.80 & 2.25 \\ 4.80 & 4.00 & 0.90 \\ 2.25 & 0.90 & 2.25 \end{bmatrix}$$

and the portfolio volatility is: $\sigma_p = \sqrt{\alpha' \Sigma \alpha} = 20.87\%$

Marginal contributions to risk are:

$$\frac{\Sigma \alpha}{\sqrt{\alpha' \Sigma \alpha}} = \frac{1}{20.87\%} \begin{bmatrix} 6.135\% \\ 3.470\% \\ 1.980\% \end{bmatrix} = \begin{bmatrix} 29.397\% \\ 16.627\% \\ 9.487\% \end{bmatrix}.$$

Risk contributions are:

$$RC = \alpha \circ \frac{\Sigma \alpha}{\sqrt{\alpha' \Sigma \alpha}} = \begin{bmatrix} 0.5 \\ 0.2 \\ 0.3 \end{bmatrix} \circ \begin{bmatrix} 29.397\% \\ 16.627\% \\ 9.487\% \end{bmatrix} = \begin{bmatrix} 14.698\% \\ 3.325\% \\ 2.846\% \end{bmatrix}.$$

The sum of risk contributions is equal to the volatility:

$$\sum_{i=1}^3 RC_i = 14.698\% + 3.325\% + 2.846\% = 20.87\% = \sigma_p.$$

Remark: An investor may want to build a portfolio which has the same RC for each asset.

Construction of Diversified Portfolios

To avoid the instability of mean-variance portfolios, some “well-diversified” portfolios have been proposed.

They have some common properties:

- the optimal portfolio does not depend on the expected returns
- it is optimally diversified according to some statistical measure.

Several well-diversified portfolios have been proposed:

- **Equally Weighted Portfolio**
- **Global Minimum Variance Portfolio**
- **Equally Weighted Risk Contribution Portfolio**
- **Most-Diversified Portfolio**

Maximum Sharpe Ratio Portfolio

The **Maximum Sharpe Ratio Portfolio** is generally not well diversified.

The objective is to maximize: $\max_{\{\alpha\}} \text{SR}_p = \frac{\bar{R}_p}{\sigma_p}$

It is not possible to further increase its risk premium without increasing its risk. This implies that for all assets the ratios of marginal contributions to risk and reward are constant. An asset's marginal contribution to portfolio risk equals $\text{MCR}_i = \partial\sigma_p/\partial\alpha_i$, whereas an asset's marginal contribution to the portfolio risk premium simply equals its risk premium, $\partial\bar{R}_p/\partial\alpha_i = \bar{R}_i$.

Therefore, for all assets within the MSRP, we must have:

$$\frac{1}{\bar{R}_i} \frac{\partial\sigma_p}{\partial\alpha_i} = \frac{1}{\bar{R}_j} \frac{\partial\sigma_p}{\partial\alpha_j} \quad \text{or equivalently} \quad \frac{\bar{R}_i}{\beta_{ip}} = \frac{\bar{R}_j}{\beta_{jp}} = \bar{R}_p$$

Equally Weighted Portfolio

The **Equally Weighted Portfolio** attributes the same weight ($\alpha_i = \alpha_j = 1/N$) to all the assets of the portfolio (naive diversification):

$$\sigma_p^2 = \frac{1}{N^2} \sum_{i=1}^N \sigma_{ii} + \frac{1}{N^2} \sum_{i=1}^N \sum_{j \neq i}^N \rho_{ij} \sigma_{ii} \sigma_{jj}$$

Properties:

- It avoids concentrated positions – in terms of money allocation
- Within equities, it implies an exposure to the small-cap anomaly
- It implies a “volatility pumping” effect: to maintain the $1/N$ allocation, we have to buy (sell) under- (out-) performing assets: “buy low, sell high” strategy
- In Bayesian terms, the $1/N$ portfolio is the “uninformed prior” (neutral diversification benchmark)

If $\rho_{ij} = \rho$ and $\sigma_{ij} = \sigma^2$, then $\sigma_p^2 = \sigma^2 \left(\rho + \frac{1}{N}(1 - \rho) \right)$

When N increases, the portfolio variance decreases to $\sigma_p^2 = \sigma^2 \rho$.

Global Minimum Variance Portfolio

The **Global Minimum Variance Portfolio** attributes the same marginal volatility to all the assets:

$$\frac{\partial \sigma_p}{\partial \alpha_i} = \frac{\partial \sigma_p}{\partial \alpha_j} = \frac{\partial \sigma_p}{\partial \alpha} \quad \text{or} \quad \text{MCR}_i = \text{MCR}_j = \text{MCR}$$

(if there were not equal, it would be possible to decrease the portfolio variance.)

Optimal weights are:

$$\alpha = \frac{\Sigma^{-1}e}{e'\Sigma^{-1}e} \quad \text{with} \quad \sigma_p^2 = \frac{1}{e'\Sigma^{-1}e}$$

Marginal volatilities are equal to the portfolio volatility:

$$\sigma_p = \sum_{i=1}^N \alpha_i \text{MCR}_i = \text{MCR} \sum_{i=1}^N \alpha_i = \text{MCR}$$

If assets are uncorrelated ($\rho_{ij} = 0$), then $\alpha_i \propto 1/\sigma_i^2$

Most-Diversified Portfolio

The **Most-Diversified Portfolio (MDP)** attributes the same relative marginal volatility to all the assets by solving

$$\max_{\{\alpha\}} \text{DR}(\alpha) = \frac{\alpha' \sigma}{\sqrt{\alpha' \Sigma \alpha}} = \frac{\sum_{i=1}^N \alpha_i \sigma_i}{\sigma_p},$$

and σ is the vector of asset volatilities. $\text{DR}(\alpha)$ is the ratio between the weighted average of volatilities and the portfolio volatility.

The MDP is the portfolio that maximizes $\text{DR}(\alpha)$. Therefore, we have the FOC condition

$$\frac{1}{\sigma_i} \frac{\partial \sigma_p}{\partial \alpha_i} = \frac{1}{\sigma_j} \frac{\partial \sigma_p}{\partial \alpha_j} \quad \text{or} \quad \frac{1}{\sigma_i} \text{MCR}_i = \frac{1}{\sigma_j} \text{MCR}_j$$

Properties:

- $\text{DR}(\alpha) = 1$ when the portfolio is invested in one asset, $\text{DR}(\alpha) > 1$ otherwise
- All the assets have the same correlation with the MDP
- If all the assets have the same Sharpe ratio, then MSRP and MDP coincide.

Reading: Choueifaty and Coignard (Journal of Portfolio Management, 2008)

Equally Weighted Risk Contribution Portfolio

The **Equally Weighted Risk Contribution Portfolio (ERC)** attributes the same risk contribution to all the assets:

$$\alpha_i \frac{\partial \sigma_p}{\partial \alpha_i} = \alpha_j \frac{\partial \sigma_p}{\partial \alpha_j} \quad \text{or} \quad \alpha_i \text{MCR}_i = \alpha_j \text{MCR}_j$$

In general, no closed-form solution.

Special cases:

- If $\rho_{ij} = \rho$, the weight of asset i is inversely proportional to its volatility:

$$\alpha_i = \frac{\sigma_i^{-1}}{\sum_{j=1}^N \sigma_j^{-1}}$$

- If $\sigma_{ii} = \sigma$, the weight of asset i is inversely proportional to average of the correlations with the other assets,

$$\alpha_i = \left(\sum_{j=1}^N \alpha_j \rho_{ij} \right)^{-1}$$

Equally Weighted Risk Contribution Portfolio

The ERC portfolio can be found by solving the problem:

$$\min_{\{\alpha\}} f(\alpha) = \sum_{i=1}^N (\sigma_i(\alpha) - \sigma_1(\alpha))^2 \quad \text{s.t.} \quad \alpha' e = 1, \quad 0 \leq \alpha_i \leq 1,$$

where $\sigma_i(\alpha) = \left(\alpha_i^2 \sigma_{ii} + \sum_{j \neq i}^N \alpha_i \alpha_j \sigma_{ij} \right) / \sigma_p$ is the risk contribution (RC) of asset i

Properties:

- $\sigma(\alpha_{GMV}) \leq \sigma(\alpha_{ERC}) \leq \sigma(\alpha_{1/N})$
- ERC portfolio is the market portfolio when $\rho_{ij} = \rho$ and all the assets have the same Sharpe ratio.

Reading: Maillard, Roncalli, and Teiletche (Journal of Portfolio Management, 2010)

Equally Weighted Risk Contribution Portfolio

The ERC portfolio can be found alternatively by solving the problem:

$$\begin{cases} \min_{\{\alpha\}} & \frac{1}{2} \alpha' \Sigma \alpha \\ \text{s.t.} & \sum_{i=1}^N \log(\alpha_i) \geq c. \end{cases}$$

The Lagrangian is: $\mathcal{L} = \frac{1}{2} \alpha' \Sigma \alpha - \lambda_c \left(\sum_{i=1}^N \log(\alpha_i) - c \right)$

The optimum gives: $\frac{\partial \mathcal{L}}{\partial \alpha_i} = 0 \quad \Rightarrow \quad (\Sigma \alpha)_i - \frac{\lambda_c}{\alpha_i} = 0$

so that $\alpha_i (\Sigma \alpha)_i = \lambda_c$

Since $\text{RC}_i = (\alpha_i (\Sigma \alpha)_i) / \sigma_p$, we have $\alpha_i (\Sigma \alpha)_i = \sigma_p \text{RC}_i$

Therefore, $\text{RC}_i = \text{RC}_j = \lambda_c / \sigma_p$, i.e. solving this objective function imposes $\text{RC}_i = \text{RC}_j$.

Inverse Volatility Portfolio

The **Inverse Volatility Portfolio (IVP, or naive risk parity)** attributes weights proportional to the stand-alone volatility of the asset:

$$\alpha_i = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}.$$

It corresponds to a volatility weighting in the cross section.

Properties:

- When volatilities are uniform, the IVP is the $1/N$ portfolio
- When all correlations are uniform (or zero), the IVP is the MVP

Condition on Risk Factors

- EW:

$$\alpha_i = \alpha_j = 1/N$$

- GMV:

$$\frac{\partial \sigma_p}{\alpha_i} = \frac{\partial \sigma_p}{\alpha_j} = \frac{\partial \sigma_p}{\alpha} \quad \text{or} \quad \text{MCR}_i = \text{MCR}_j = \text{MCR}$$

- MDP:

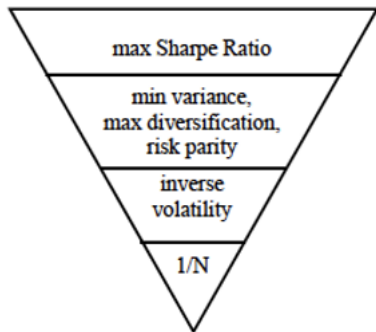
$$\frac{1}{\sigma_i} \frac{\partial \sigma_p}{\alpha_i} = \frac{1}{\sigma_j} \frac{\partial \sigma_p}{\alpha_j} \quad \text{or} \quad \frac{\text{MCR}_i}{\sigma_i} = \frac{\text{MCR}_j}{\sigma_j}$$

- ERC:

$$\alpha_i \frac{\partial \sigma_p}{\alpha_i} = \alpha_j \frac{\partial \sigma_p}{\alpha_j} \quad \text{or} \quad \alpha_i \text{MCR}_i = \alpha_j \text{MCR}_j$$

Condition on Risk Factors

The portfolio decision pyramid: link between portfolio rules and the information burden placed on the investor when adopting that rule.



4. full information on risk premia, standard deviations and correlations
3. full information on standard deviations and correlations
2. full information on standard deviations
1. no meaningful differences among risk premia and risk inputs

Risk Budgeting Portfolio

A risk budgeting portfolio α satisfies the following conditions:

$$\left\{ \begin{array}{l} \text{RC}_1 = b_1 R(\alpha) \\ \vdots \\ \text{RC}_i = b_i R(\alpha) \\ \vdots \\ \text{RC}_N = b_N R(\alpha) \end{array} \right.$$

where $R(\alpha)$ is a risk measure (for instance, the portfolio volatility) and $b = (b_1, \dots, b_N)$ is a vector of risk budgets such that $b_i \geq 0$ and $\sum_{i=1}^N b_i = 1$.

The **ERC portfolio** is a particular case of a risk budgeting portfolio when $R(\alpha) = \sigma_p(\alpha)$ and $b_i = 1/N$.

End of Lecture 3